

1 What is a Process?

A process is a program in execution.

While a program is a passive entity stored on disk, a process is an active entity that includes:

- Program code (text section)
- Current activity (Program Counter)
- CPU registers
- Stack (function calls, local variables)
- Data section (global variables)
- Heap (dynamic memory)
- OS-managed resources (files, I/O devices, etc.)

👉 In simple words:

Program = Recipe

Process = Cooking activity

2 Process vs Program

Program	Process
Static (stored on disk)	Dynamic (executing)
Passive entity	Active entity
No state	Has states
No resources allocated	Resources allocated

3 Components of a Process

A process in memory typically contains:

1. Text Section → Executable instructions
2. Data Section → Global/static variables
3. Heap → Dynamically allocated memory
4. Stack → Function calls, parameters, local variables
5. Registers → CPU state (PC, SP, etc.)

4 Process Control Block (PCB)

The PCB is a data structure maintained by the OS to manage processes.

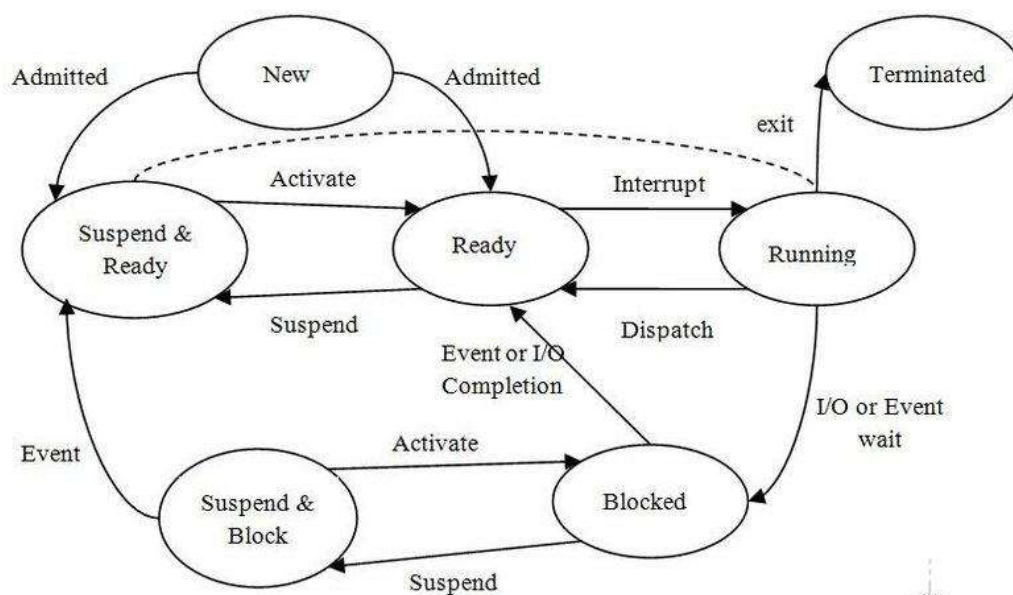
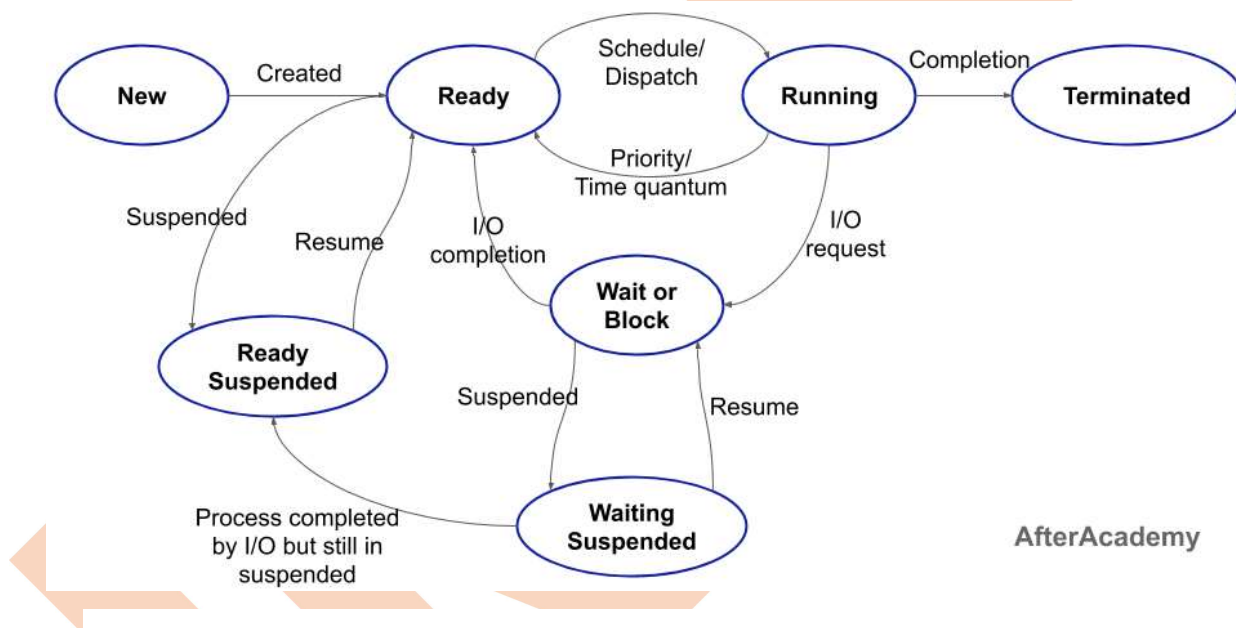
Information stored in PCB:

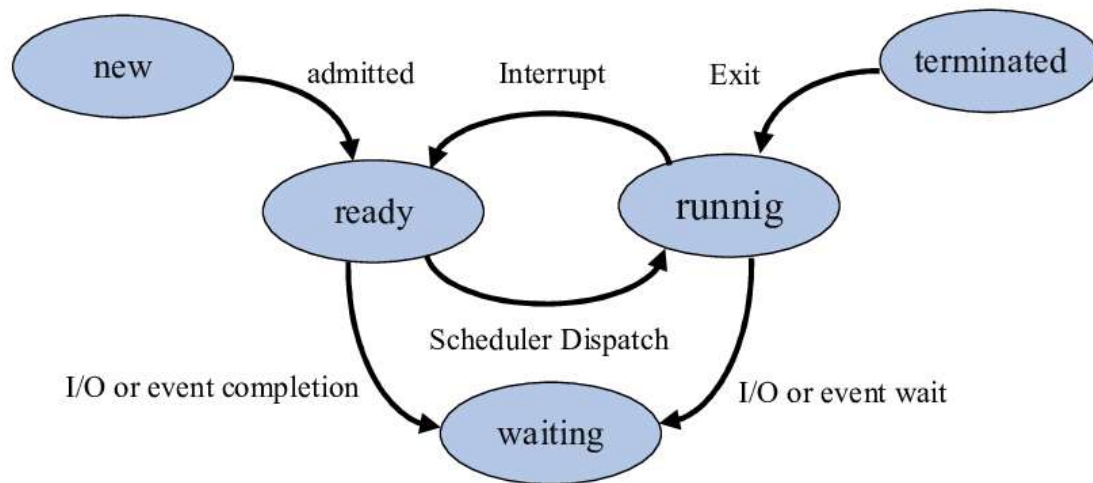
- Process ID (PID)

- **Process State (new, ready, running...)**
 - **Program Counter**
 - **CPU Registers**
 - **CPU Scheduling Info (priority, queue pointers)**
 - **Memory Management Info (base/limit registers, page tables)**
 - **Accounting Info (CPU time used, user ID)**
 - **I/O Status Info (open files, devices)**
- 👉 **PCB = Identity card + medical record + timetable of a process**

5 Process States

A process moves through different states during execution.





4 Common 5-State Model:

1. New → Process is being created
2. Ready → Waiting for CPU
3. Running → Currently executing
4. Waiting / Blocked → Waiting for I/O/event
5. Terminated → Execution finished

6 State Transitions

Transition	Reason
New → Ready	Admitted by OS
Ready → Running	CPU allocated
Running → Ready	Time slice expired
Running → Waiting	I/O request
Waiting → Ready	I/O completed
Running → Terminated	Process finished

7 Process Scheduling

The OS decides:

- Which process runs?
- For how long?
- In what order?

Queues involved:

- Job Queue → All processes in system
- Ready Queue → Processes waiting for CPU
- Device Queue → Processes waiting for I/O

Schedulers:

1. **Long-Term Scheduler** → Controls degree of multiprogramming
2. **Short-Term Scheduler (CPU Scheduler)** → Selects next process
3. **Medium-Term Scheduler** → Swapping/suspending

8 Context Switching

When CPU switches from one process to another:

- Save current process state in PCB
- Load next process state from PCB
- 🕒 **Overhead** → No useful work done during switch
- 👉 **Efficient OS minimizes context switches**

9 Process Operations

Process Creation

Occurs via:

- System initialization
- User request
- Batch job submission
- Process spawning another process

Steps:

1. **Assign PID**
2. **Create PCB**
3. **Allocate memory/resources**
4. **Add to ready queue**

10 Process Termination

Reasons:

- Normal completion
- Error exit
- Fatal exception
- Killed by another process/OS

After termination:

- Resources released
- PCB deleted

1 1 Parent & Child Processes

- Processes can create other processes
- Parent-child relationship

Execution possibilities:

- Parent & child run concurrently

- Parent waits for child

1 2 Interprocess Communication (IPC)

Processes may cooperate:

Two IPC Models:

1. Shared Memory

- Fast
- Requires synchronization

2. Message Passing

- Send/receive messages
- Safer but slower

1 3 Types of Processes

- CPU-bound → Heavy computation
- I/O-bound → Frequent I/O operations
- Foreground / Interactive
- Background / Batch
- Real-time Processes

1 4 Process vs Thread

Process	Thread
Heavyweight	Lightweight
Separate memory	Shared memory
Expensive switch	Cheap switch
Independent	Dependent

👉 Threads are units of execution within a process.

1 5 Key Concepts Summary

- ✓ Process = Program in execution
- ✓ PCB = Core management structure
- ✓ States = Lifecycle stages
- ✓ Scheduling = CPU allocation strategy
- ✓ Context Switch = State saving/loading
- ✓ IPC = Process cooperation mechanism

Practice Sheet

01. A process stack does NOT contain:
(a) Function parameters
(b) Local variables (c) Return addresses
(d) PID (Process Identifier)
02. Which transition is NOT possible in a standard process state model?
(a) Ready → Running
(b) Running → Ready
(c) Waiting → Running
(d) Running → Waiting
03. In the PCB, where is the address of the next instruction to be executed stored?
(a) Process Stack (b) Program Counter
(c) Register Set (d) Process Status Word
04. What is the main objective of Multi-programming?
(a) To minimize CPU utilization
(b) To maximize CPU utilization
(c) To provide interactive computing
(d) To decrease the number of processes in the ready queue
05. The “Context” of a process includes:
(a) Only the CPU registers
(b) Only the program counter
(c) The PCB, stack, and address space
(d) Only the user data
06. Which of the following is a non-preemptive scheduling algorithm?
(a) Round Robin
(b) Priority Scheduling (Preemptive)
(c) FCFS (First-Come, First-Served)
(d) Shortest Remaining Time First (SRTF)
07. In a Round Robin scheduling, as the time quantum increases, the performance approaches:
(a) Shortest Job First (b) FCFS
(c) Priority Scheduling
(d) Multi-level Queue Scheduling
08. Which of the following CPU scheduling algorithms results in the minimum average waiting time?
(a) FCFS
(b) SJF (Shortest Job First)
(c) Round Robin
(d) Priority Scheduling
09. “Starvation” is a primary risk in which algorithm?
(a) FCFS (b) Round Robin
(c) Priority Scheduling (d) All of the above
10. A process that has finished execution but whose parent has not yet called wait() is:
(a) Orphan (b) Zombie
(c) Daemon (d) Trap
11. The interval from the time of submission of a process to the time of completion is:
(a) Waiting Time (b) Response Time
(c) Throughput (d) Turnaround Time
12. Which scheduler is responsible for swapping processes in and out of the disk?
(a) Short-term (b) Long-term
(c) Medium-term (d) Dispatcher
13. In the context of threads, “User-level threads” are:
(a) Managed by the Kernel
(b) Managed by a thread library
(c) Slower to create than kernel threads
(d) Supported directly by the OS
14. What is the main disadvantage of the Shortest Job Next (SJN) algorithm?
(a) High overhead
(b) Difficulty in knowing the length of the next CPU burst
(c) Low throughput
(d) It is always preemptive
15. A “Race Condition” occurs when:
(a) Multiple processes access and manipulate the same data concurrently
(b) A process stays in the ready queue for too long
(c) The CPU is idle for too long
(d) Two processes are in a deadlock
16. The section of code where shared resources are accessed is called:
(a) Entry Section (b) Critical Section
(c) Exit Section (d) Remainder Section
17. Which of the following is a hardware-based solution to the critical section problem?
(a) Peterson’s Solution (b) Semaphores
(c) Test-and-Set instruction (d) Monitors

18. A binary semaphore can take values:
(a) 0 and 1
(b) Any integer
(c) Only positive integers
(d) -1 and 1
19. What happens when a process performs a P (or wait) operation on a semaphore with value 0?
(a) The value becomes -1
(b) The process is blocked
(c) The process enters the critical section
(d) The value remains 0 and the process continues
20. Which of the following is an example of an IPC (Inter-Process Communication) mechanism?
(a) Shared Memory (b) Message Passing
(c) Pipes (d) All of the above
21. A "Context Switch" is performed by the:
(a) Long-term Scheduler
(b) Medium-term Scheduler
(c) Dispatcher
(d) Shell
22. In Multi-level Feedback Queue scheduling, processes are moved between queues based on:
(a) Their PID
(b) Their CPU burst characteristics
(c) User preference
(d) Memory requirements
23. Which of the following is TRUE about threads?
(a) Threads share the stack of the parent process
(b) Threads share the code and data section of the parent process
(c) Threads have their own unique PID
(d) Switching between threads is slower than switching between processes
24. The "Convoy Effect" is a major drawback of:
(a) SJF
(b) Round Robin
(c) FCFS
(d) Priority Scheduling
25. If the time quantum in Round Robin is very small, it leads to:
(a) Very high throughput
(b) High context-switching overhead
(c) No starvation
(d) Better CPU utilization
26. The fork() system call returns _____ to the parent process on success.
(a) 0
(b) -1
(c) PID of the child process
(d) PID of the parent process
27. In a "Many-to-One" thread model:
(a) Many kernel threads map to one user thread
(b) Many user-level threads map to one kernel thread
(c) One user thread can run on multiple CPUs
(d) Blocking one thread does not block others
28. Which of the following is NOT a necessary condition for Deadlock?
(a) Mutual Exclusion (b) Hold and Wait
(c) Preemption (d) Circular Wait
29. The "Banker's Algorithm" is used for:
(a) Deadlock Detection
(b) Deadlock Prevention
(c) Deadlock Avoidance
(d) Deadlock Recovery
30. A process is "I/O Bound" if it:
(a) Spends more time doing computations
(b) Spends more time doing I/O requests
(c) Uses very little memory
(d) Always runs in the background
31. The "Degree of Multiprogramming" refers to:
(a) The number of CPUs in the system
(b) The number of processes in the ready queue
(c) The number of processes in memory
(d) The number of threads per process
32. What is a "Signal" in Unix-like systems?
(a) A way to share memory
(b) A software interrupt to notify a process of an event
(c) A hardware error
(d) A scheduling priority

33. In the “Dining Philosophers” problem, the chopsticks represent:
- Processes
 - Shared Resources (Semaphores)
 - CPU cycles
 - Users
34. The “Sleeping Barber” problem is a classic example of:
- Process Synchronization
 - Memory Management
 - Deadlock Avoidance
 - Disk Scheduling
35. A “Pipe” in OS is used for:
- One-way communication between related processes
 - Two-way communication between unrelated processes
 - Increasing CPU speed
 - Managing the process stack
36. “Aging” is a technique used to:
- Kill old processes
 - Solve the problem of starvation in priority scheduling
 - Increase the time quantum of Round Robin
 - Reduce the degree of multiprogramming
37. Which state transition occurs when a process finishes its I/O operation?
- Running → Ready
 - Waiting → Ready
 - Waiting → Running
 - Ready → Running
38. The exec() system call:
- Creates a new process
 - Replaces the current process image with a new program
 - Terminates the process
 - Suspends the process
39. “Throughput” is defined as:
- Number of processes completed per unit time
 - Total time taken to execute a job
 - Time spent in the ready queue
 - Amount of memory used by a process
40. A “Lightweight Process” (LWP) is another name for:
- A process with low priority
 - A Thread
 - A Daemon
 - A Zombie
41. In the Bounded-Buffer problem, which semaphore is used to track the number of empty slots?
- Mutex
 - Full
 - Empty
 - Binary
42. Which of the following is a “Preemptive” version of SJF?
- FCFS
 - SRTF (Shortest Remaining Time First)
 - Priority Scheduling
 - Round Robin
43. A “Critical Section” must satisfy which three requirements?
- Mutual Exclusion, Progress, Bounded Waiting
 - Mutual Exclusion, Speed, Progress
 - Atomicity, Consistency, Isolation
 - Hold and Wait, No Preemption, Circular Wait
44. What is “Priority Inversion”?
- When a low priority process runs before a high priority one
 - When a high priority process is delayed by a lower priority one holding a resource
 - Changing process priority over time
 - The dispatcher failing to switch processes
45. The “Ready Queue” is usually implemented as a:
- Stack
 - Linked List
 - Tree
 - Hash Table
46. Two processes P_1 and P_2 share a variable x (initially 0). They execute $x = x + 1$ and $x = x - 1$ respectively. If these execute concurrently without synchronization, the final value of x can be:
- Only 0
 - 1, 0, or 1
 - Any integer
 - Always 1

47. In the Producer-Consumer problem with a buffer of size N, if we swap the order of wait(empty) and wait(mutex) in the Producer, what happens?
- (a) The program runs faster.
 - (b) A deadlock can occur if the buffer is full.
 - (c) Mutual exclusion is violated.
 - (d) Nothing changes; it's logically equivalent.
48. A monitor is a programming language construct that provides:
- (a) Lower-level synchronization than semaphores.
 - (b) Automatic mutual exclusion for its procedures.
 - (c) Faster context switching than threads.
 - (d) A way to bypass the kernel.
49. Consider three processes P₁, P₂, P₃ with priorities 10, 20, 30 (higher value = higher priority). P₁ holds a lock that P₃ needs. P₂ is ready to run. If P₂ preempts P₁, this is called:
- (a) Priority Inheritance.
 - (b) Priority Inversion.
 - (c) Deadlock.
 - (d) Convoy Effect.
50. Which of the following is true about "Spinlocks"?
- (a) They are ideal for single-processor systems.
 - (b) They avoid the overhead of context switching.
 - (c) They move the process to a waiting queue.
 - (d) They are useful only for long-duration tasks.

- | | |
|---|---------|
| 01. (d)
The PID is stored in the PCB (Process Control Block) data section, not the process stack. The stack is for temporary data like function calls and local variables. | 19. (b) |
| 02. (c)
A process cannot go directly from Waiting to Running. It must first go to the Ready queue and wait for the CPU scheduler to pick it. | 20. (d) |
| 03. (b)
The Program Counter (PC) specifically holds the memory address of the next instruction to be fetched. | 21. (c) |
| 04. (b)
Multiprogramming keeps the CPU busy at all times by having multiple processes in memory ready to run. | 22. (b) |
| 05. (c) | 23. (b) |
| 06. (c) | 24. (c) |
| 07. (b) | 25. (b) |
| 08. (b) | 26. (c) |
| 09. (c) | 27. (b) |
| 10. (b) | 28. (c) |
| 11. (d) | 29. (c) |
| 12. (c) | 30. (b) |
| 13. (b) | 31. (c) |
| 14. (b) | 32. (b) |
| 15. (a) | 33. (b) |
| 16. (b) | 34. (a) |
| 17. (c) | 35. (a) |
| 18. (a) | 36. (b) |
| | 37. (b) |
| | 38. (b) |
| | 39. (a) |
| | 40. (b) |
| | 41. (c) |
| | 42. (b) |
| | 43. (a) |
| | 44. (b) |
| | 45. (b) |
| | 46. (b) |
| | 47. (b) |
| | 48. (b) |
| | 49. (b) |
| | 50. (b) |